

Bruno Ferreira

AI & Software Engineer

Machine Learning · Data Science · LLMs · Automation

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I build applications, integrations and automations, connecting process analysis, data and software engineering. I combine Python, Machine Learning and LLMs with NLP research and solution evaluation. Client engineering since 2025 and university teaching since 2024, with an MSc in Informatics Engineering.

PROFESSIONAL EXPERIENCE

Applied AI Engineer and Consultant

Independent client work · Portugal and remote

Jun 2025–present

- Develop solutions from workflow analysis and requirements to implementation, testing and handover documentation, within each agreed client scope.
- Advise on AI use cases and acceptance criteria; design and deliver practical training for client and developer teams.

SELECTED CLIENT PROJECTS

HVAC specification extraction

Implemented and evaluated

Jan–Mar 2026

Deliverable. Built a Python pipeline that extracts PDF specifications into Excel and JSON, retaining the value, source page and extraction method.

Validation. Compared fields against manual references; implemented tests for equipment attribution, identifier collisions and insufficient retrieval context.

Python, LlamaIndex, ChromaDB, RAG, PDF/tables, vision, Excel, JSON · [Case study](#)

LegalTech platform for US visa workflows

Components in development

2025–present

Application components. Designed and implemented triage, dynamic questionnaires and document management to structure case information in a multi-tenant application.

Review output. Combined OCR and LLM analysis in evidence-guided reports for review against USCIS criteria.

Python, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs

Quality Copilot for packaging and technical documents

Private pilot

2026

Scope. Built a six-stage pilot covering document comparison, artwork dimensions and content, corrections, supplier proofs and technical-sheet preparation.

Deliverable. Produced stage-specific reports with differences and supporting evidence for human validation.

Python, OCR, Claude Code skills, document automation

TECHNICAL SKILLS

- Machine Learning:** Classical ML and predictive modelling
- Data Science:** Data preparation, cleaning, preprocessing, exploratory analysis and visualisation
- AI engineering:** LLMs, RAG, embeddings, NLP, OCR, information extraction, evaluation
- Software:** Python, FastAPI, TypeScript, React, Next.js, SQL, PostgreSQL, Docker, Git
- Delivery:** Process discovery, requirements, API integration, testing, documentation, training

TEACHING AND RESEARCH

Invited Lecturer

- University of Coimbra
- Feb 2024–present
- Teach databases, AI, Java, C, natural language processing and knowledge representation.
 - Prepare learning materials, assess student projects and supervise MSc dissertations.

Researcher

- CISUC · Centre for Responsible AI
- Feb 2022–Jun 2025
- Researched NLP, information extraction, question answering and the evaluation of knowledge encoded in language models.
 - Published work on language-model evaluation and document information extraction, alongside research in explainable AI, causal inference and model compression.

EDUCATION

PhD in Informatics Engineering

- University of Coimbra · In progress
- 2022–present

MSc in Informatics Engineering — Intelligent Systems

- University of Coimbra
- 2022

BSc in Informatics Engineering

- University of Coimbra
- 2021

ADDITIONAL WORK

Privacy-by-design AI platform

- AI and privacy
- 2026
- Built a functional React/TypeScript PWA using local NER and ONNX for personal-data detection and anonymisation before external LLM workflows.

THE 17.3 evidence evaluation

- Commissioned project
- 2025
- Designed and carried out a repeatable review of public university evidence across 17 THE Impact Rankings indicators.

SELECTED PUBLICATIONS

- Towards Automated Evaluation of Knowledge Encoded in Large Language Models · DLnLD @ LREC-COLING 2024
- BATS-PT: Assessing Portuguese Masked Language Models in Lexico-Semantic Analogy Solving and Relation Completion · PROPOR 2024
- Leveraging Question Answering for Domain-Agnostic Information Extraction · CIARP 2023
- [Full publication list and project details](#)

LANGUAGES AND AVAILABILITY

- Portuguese: native · English: advanced · Spanish: good comprehension, basic speaking
- Based in Pombal, Portugal. Open to AI, software and data roles, as well as B2B projects and ongoing engineering support through my company.